

# Manoel Da Ponte

AI ENGINEER • DATA ENGINEER

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AI and data engineer, co-founder of WiseTwin, where I designed and shipped LLM agents, RAG pipelines and a multitenant SaaS platform. Previously data engineer at TotalEnergies in Denmark and data scientist at CGI. French citizen (EU/EFTA national), based in Switzerland.

## Skills

### Generative AI & Agents

LLM agents (tool use), RAG, pgvector, LLM-as-judge, SSE, multimodal vision

### Data Engineering

Idempotent ETL, PostgreSQL, pgvector, SQLAlchemy, Prisma, pandas, Prefect

### Machine Learning

NLP, computer vision, LLM fine-tuning, deep learning, time series

### Backend

Python, FastAPI, Node.js, REST APIs, Pydantic, Zod

### Frontend

Next.js, React, TypeScript, Tailwind CSS

### Cloud & DevOps

Azure (Container Apps, Blob, ACR), Docker, GitHub Actions, Vercel, AWS

### Real-Time 3D

PlayCanvas, Gaussian Splatting, GLSL shaders, Unity, Google Maps 3D

## Work Experience

### WiseTwin [↗](#)

Dunkirk, France (remote)

CO-FOUNDER & LEAD DEVELOPER

March 2024 - Present

- Designed and built a multitenant SaaS platform for industrial safety training, spanning three products (immersive 3D training, document-based courses, prevention tours): 80 000 lines of TypeScript, 124 API routes
- Built an HSE AI copilot: LLM agent (Claude) with 12 tools, agentic loop implemented without a framework, SSE streaming, sensitive actions gated by human confirmation
- Production RAG pipeline: embeddings and pgvector similarity search over 62 726 industrial accidents from the French public ARIA database, with tenant isolation enforced by CI tests
- Automated course generation from PowerPoint: Whisper transcription, multimodal LLM writing, quiz questions verified by a second model (LLM-as-judge)
- Web-based 3D editors in production: Gaussian Splatting (PlayCanvas, GPU selection via custom GLSL shaders) and photorealistic 3D maps (Google Maps 3D, geolocated GLB models)
- Security and compliance: enterprise SSO, TOTP MFA, xAPI/cmi5 and Qualiopi exports; per-organization AI cost governance (token tracking, budgets, quotas)
- Infrastructure: Azure Container Apps, Container Registry and Blob Storage, GitHub Actions CI/CD, Vercel, Neon PostgreSQL
- Technologies: Python, FastAPI, TypeScript, Next.js/React, Claude (Anthropic), OpenAI, pgvector, PostgreSQL, Prisma, Docker, Azure

### TotalEnergies

Esbjerg, Denmark

DATA ENGINEER

March 2022 - February 2024

- Delivered end-to-end data projects in small teams (1 to 3 people), covering data management, backend and frontend
- Accident prevention solution: IoT data extraction, NLP analysis of incident reports, centralized data warehouse and LLM fine-tuning
- Invoice control system combining computer vision and predictive analysis for procurement optimization
- Offshore IoT architecture: data warehouse, anomaly detection, Prefect monitoring and automated alerting
- Lightweight tools for large-scale SAP data extraction used by business teams
- Application deployments on Azure and on-premise through CI/CD pipelines
- Technologies: Python, Azure, SQL, NLP, Computer Vision, IoT, Prefect, SAP, React

### CGI

Toulouse, France

DATA SCIENTIST

July 2020 - March 2022

- Chatbot for technical documentation access, built on indexing and semantic search (Elasticsearch)
- Cross-system data migration solution on AWS
- Behavioral analytics and personalized recommendations to improve user journeys
- Technologies: Python, AWS, NLP, Elasticsearch, SQL, REST APIs

# Education

## MSc in Data Science and Engineering

GRADUATE

In-depth data engineering curriculum covering collection, storage and statistical analysis of large-scale data.

Paul Sabatier University, Toulouse III

2019 - 2021

## BSc in Applied Mathematics (MIASHS)

GRADUATE

Multidisciplinary curriculum combining applied mathematics, computer science and social sciences.

Jean Jaurès University, Toulouse II

2018 - 2019

## Two-year degree (DUT) in Statistics and Business Intelligence

GRADUATE

Statistics and computer science applied to data analysis and decision-support modeling.

University of Caen

2016 - 2018

# Selected Projects

## WiseAtlas

DESIGNER & DEVELOPER

- Multitenant SaaS editor for interactive territory maps: photorealistic 3D mapping (Google Maps 3D), geolocated 3D models, timeline-based storytelling and rich content blocks
- Production LLM translation feature: batching, rate-limit handling, per-model cost tracking
- Technologies: Next.js, React, TypeScript, Prisma, PostgreSQL, Azure Blob, Claude (Anthropic)

WiseTwin

2025 - 2026

## Splat Editor SafetyTour

DESIGNER & DEVELOPER

- Web-based 3D Gaussian Splat editor: cleanup of digitized industrial scenes, GPU selection through GLSL shaders, educational hotspots with quizzes, guided tours and standalone HTML export
- SaaS integration: JWT authentication, Azure storage with signed URLs, scene compression, learner progress analytics
- Technologies: PlayCanvas, WebGL2, GLSL, Next.js, TypeScript, Zustand

WiseTwin

2025 - 2026

## Bootstrap-Now

CO-FOUNDER

- SaaS platform supporting new entrepreneurs: business opportunity discovery through multi-channel data aggregation and AI-assisted business plan generation
- Technologies: Next.js, Node.js, PostgreSQL, Prisma, OpenAI API, Vercel

Toulouse, France

2024 - 2025

# Languages

French

Native

English

C1 (full professional proficiency, two years in an international environment in Denmark)

# Interests

Game development

Personal Unity and C# prototypes, including a multiplayer game (Invasion)

Guitar

Playing since age 12

# References

Available upon request.